# Topic #:-

# Strassen's matrix multiplication

## Problem statement:-

Two matrices A and B and calculate matrix C by multiplying X and Y.

Time complexity using:

Conventional method: $O(n^3)$

Divide and conquer method: $O(n^3)$

=> Strassen's method is used to make the recursion tree slightly less bushy.

=> Instead of performing eight recursive multiplications of $n/2 \times n/2$ matrices, it performs only seven and 18 (+, -)

=> it has four steps.

=> Strassen's matrix multiplication can be performed only on square matrices where n is power of 2.

=> Order of both of the matrices are $n \times n$.

## Method:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \begin{bmatrix} B_{11} & B_{12} \\ B_{21} & B_{22} \end{bmatrix}$$

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$$P = (A_{11} + A_{22}) \times (B_{11} + B_{22})$$
$$Q = B_{11}(A_{21} + A_{22})$$
$$R = A_{11}(B_{12} - B_{22})$$
$$S = A_{22}(B_{21} - B_{11})$$
$$T = B_{22}(A_{11} + A_{12})$$
$$U = (B_{11} + B_{12})(A_{21} - A_{11})$$
$$V = (B_{21} + B_{22})(A_{12} - A_{22})$$

TRICK

$B_{11}$ $A_{11}$ A B

$$\begin{bmatrix} 11 & 12 \\ 21 & 22 \end{bmatrix}$$

S C

P = sum of diag

$$C_{11} = P + S - T + V$$ (P-Photo, T-tele, S-shop, V-Visio)

$$C_{12} = R + T$$ (Rat)

$$C_{21} = Q + S$$ (QS)

$$C_{22} = P + R - Q + U$$ (PRQU)

## Time Complexity:

$$\Rightarrow T(n) = \begin{cases} \Theta(1) & \text{if } n = 1 \\ 7T(n/2) + \Theta(n^2) & \text{if } n > 1 \end{cases}$$

$$\therefore T(n) = \Theta(n^{\log 7}) = \Theta(n^{2.[illegible]})$$

**Example:**

Use Strassen's algorithm to compute the matrix product

$$\begin{pmatrix} 1 & 3 \\ 7 & 5 \end{pmatrix} \begin{pmatrix} 6 & 8 \\ 4 & 2 \end{pmatrix}$$

$$A = \begin{bmatrix} 1 & 3 \\ 7 & 5 \end{bmatrix}$$

$$B = \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

$$C = \begin{bmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{bmatrix}$$

$A_{11} = 1$ , $B_{11} = 6$

$A_{12} = 3$ , $B_{12} = 8$

$A_{21} = 7$ , $B_{21} = 4$

$A_{22} = 5$ , $B_{22} = 2$

$P = (A_{11} + A_{22}) \times (B_{11} + B_{22}) = (1+5) \times (6+2) = 6 \times 8 = 48$

$Q = B_{11}(A_{21} + A_{22}) = 6(7+5) = 72$

$R = 1(8-2) = 6$

$S = 5(4-6) = -10$

$T = 2(1+3) = 8$

$U = (6+8)(7-1) = 14 \times \cancel{6} = 84$
$V = (4+2)(3-5) = 6(-2) = -12$

$C_{11} = 48 - 10 - 8 + 18 = 18$.
$C_{12} = 6 + 8 = 14$
$C_{21} = 72 - 10 = 62$
$C_{22} = 48 + 6 - 72 + 84 = 66$

$$C = \begin{bmatrix} 18 & 14 \\ 62 & 66 \end{bmatrix}$$

verification:

$$= \begin{bmatrix} 1 & 3 \\ 7 & 5 \end{bmatrix} \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 18 & 14 \\ 62 & 66 \end{bmatrix}$$

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# Write Pseudocode for Strassen's algorithm.

STRASSEN (M, N)
n = M.rows
if n == 1
return M11 N11 or C11 = a11 . b11
else

Let $S_1, S_2, \ldots$ and $S_{10}$ be 10 new $n/2 \times n/2$ matrices.

Let $P_1, P_2, \ldots$, and $P_7$ be 7 new $\frac{n}{2} \times \frac{n}{2}$ matrices

// calculate the sum

$S_1 = B_{12} - B_{22}$

$S_2 = A_{11} + A_{12}$

$S_3 = A_{21} + A_{22}$

$S_4 = B_{21} - B_{11}$

$S_5 = A_{11} + A_{22}$

$S_6 = B_{11} + B_{22}$

$S_7 = A_{12} - A_{22}$

$S_8 = B_{21} + B_{22}$

$S_9 = A_{11} - A_{21}$

$S_{10} = B_{11} + B_{12}$

// Calculate the product matrices

$P_1$ = STRassen ($A_{11}$, $S_1$)

$P_2$ = STRassen ($S_2$, $B_{22}$)

$P_3$ = STRassen ($S_3$, $B_{11}$)

$P_4$ = STRassen ($A_{22}$, $S_4$)

$P_5$ = STRassen ($S_5$, $S_6$)

$P_6$ = STRassen ($S_7$, $S_8$)

$P_7$ = STRassen ($S_9$, $S_{10}$)

// Calculate the final product

$C_{11} = P_4 + P_5 + P_6 - P_2$

$C_{12} = P_1 + P_2$

$C_{21} = P_3 + P_4$

$C_{22} = P_1 + P_5 - P_3 - P_7$

return C